MACHINE LEARNING

Polynomial regression

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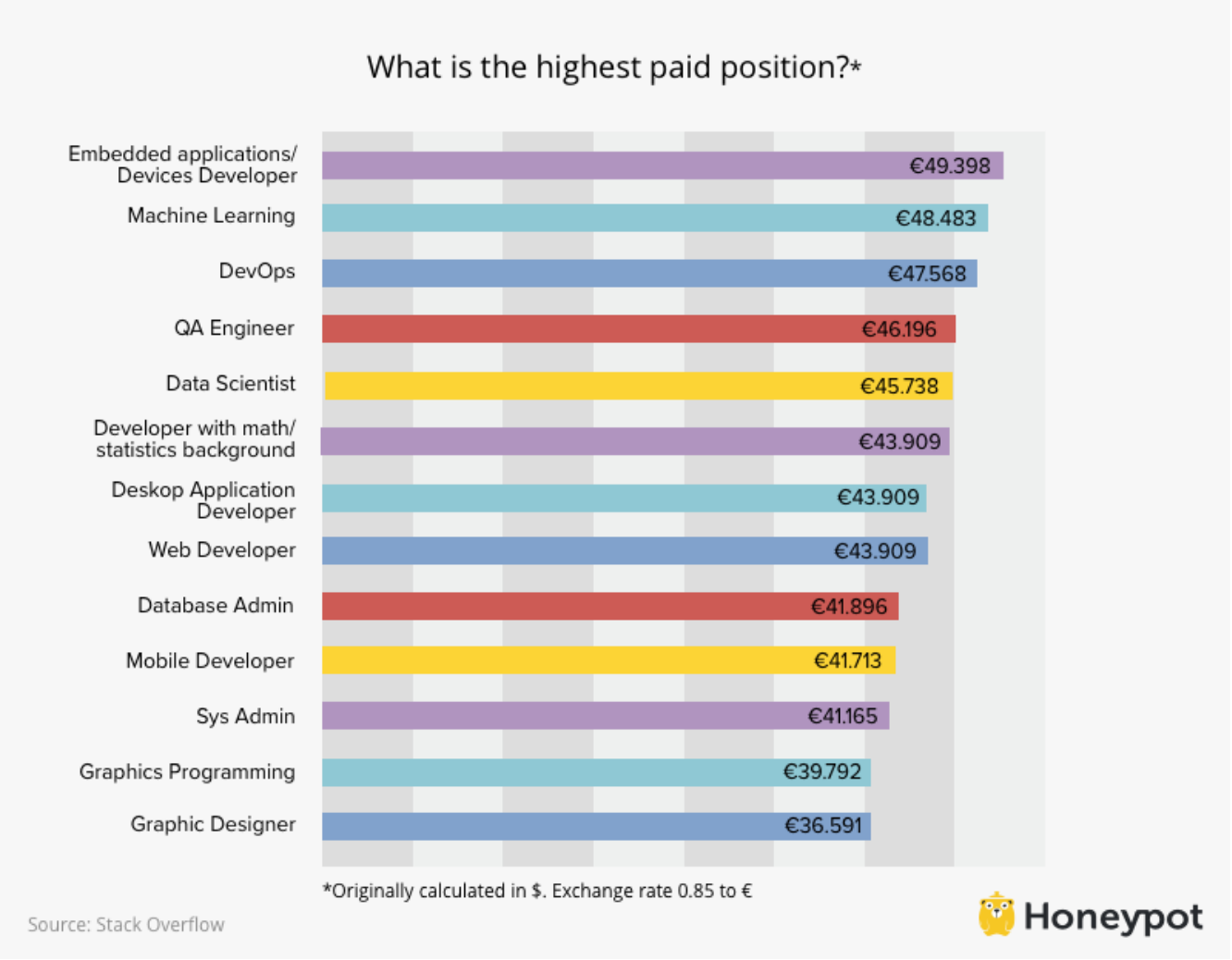
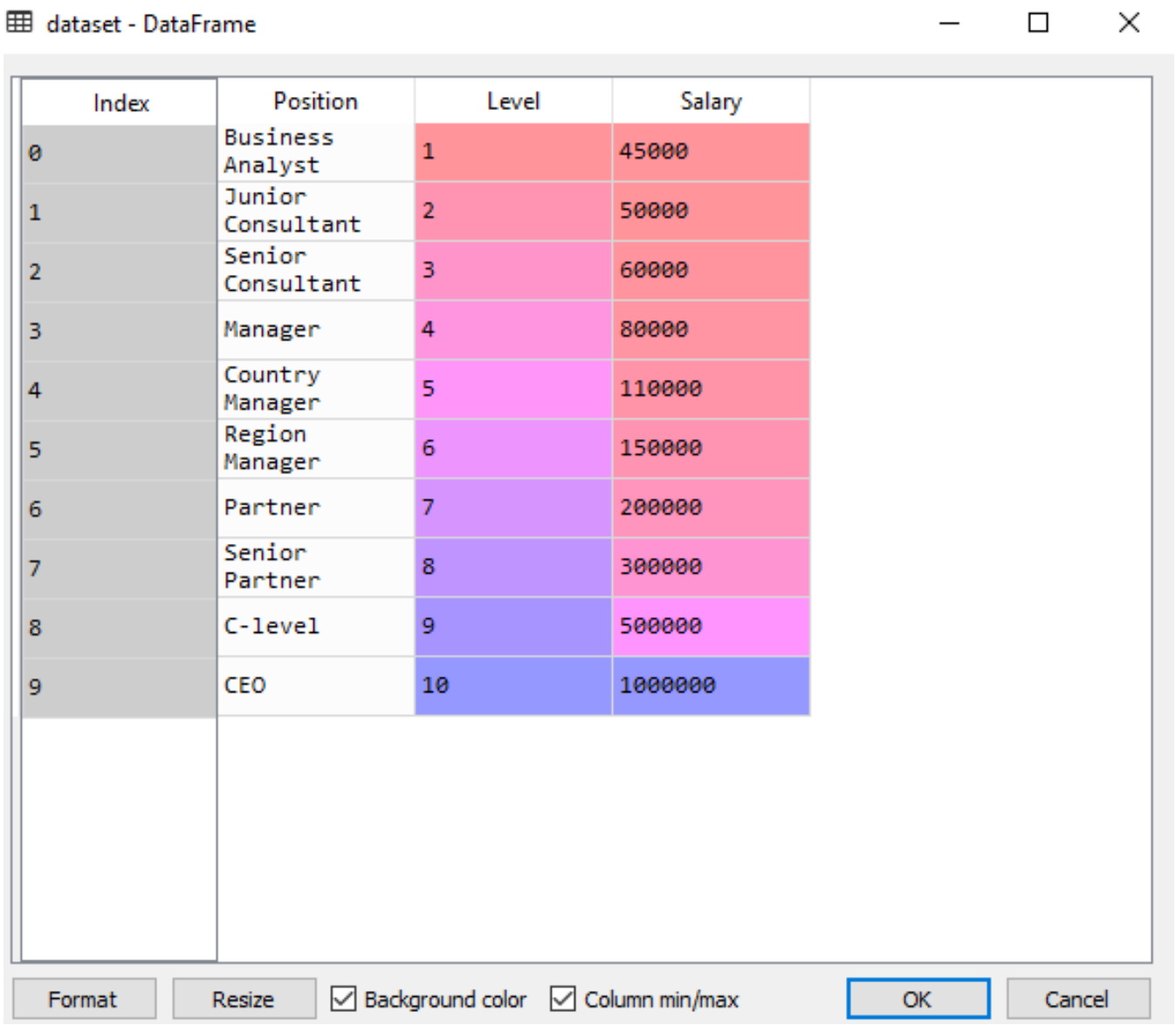
CODE EXAMPLE – POSITION SALARIES



LOAD DATASET

```
# Importing the libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Importing the dataset
dataset = pd.read_csv('Position_Salaries.csv')
```



Index	Position	Level	Salary
0	Business Analyst	1	45000
1	Junior Consultant	2	50000
2	Senior Consultant	3	60000
3	Manager	4	80000
4	Country Manager	5	110000
5	Region Manager	6	150000
6	Partner	7	200000
7	Senior Partner	8	300000
8	C-level	9	500000
9	CEO	10	1000000

DEPENDENT & INDEPENDENT VARIABLES

X = dataset.iloc[:, 1:2].values

y = dataset.iloc[:, 2].values

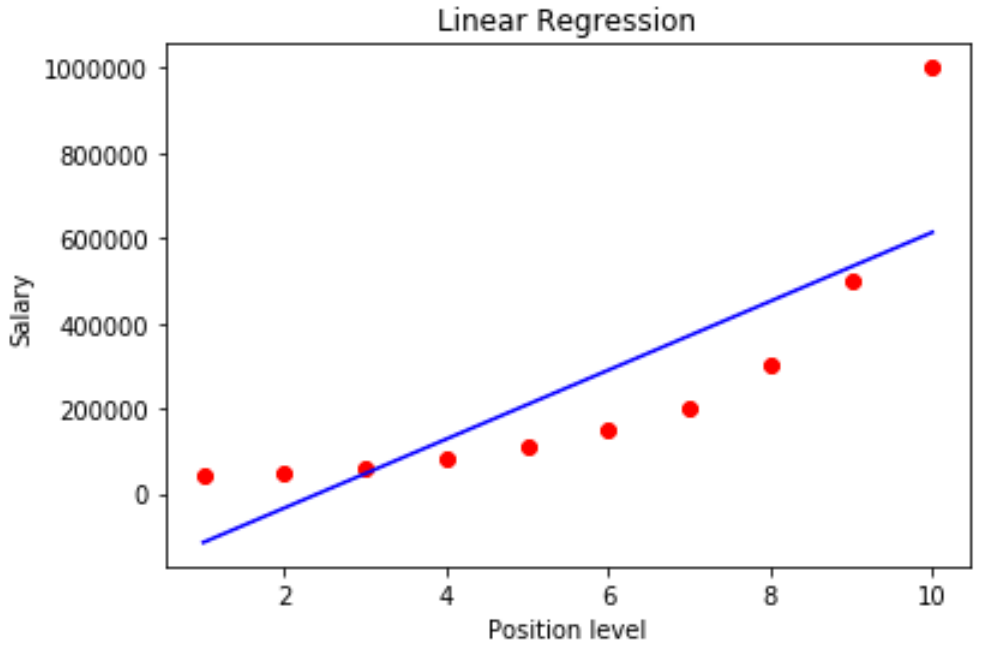
TRAINING AND TESTING SETS

Not enough Data….

FITTING LINEAR REGRESSION TO THE DATA

```
from sklearn.linear_model import LinearRegression
lin_reg = LinearRegression()
lin_reg.fit(X, y)
#we want to compare Linear regression model to polynomial regression
```

```
plt.scatter(X, y, color = 'red')
plt.plot(X, lin_reg.predict(X), color = 'blue')
plt.title('Linear Regression')
plt.xlabel('Position level')
plt.ylabel('Salary')
plt.show()
```



VISUALIZING THE LINEAR REGRESSION RESULTS

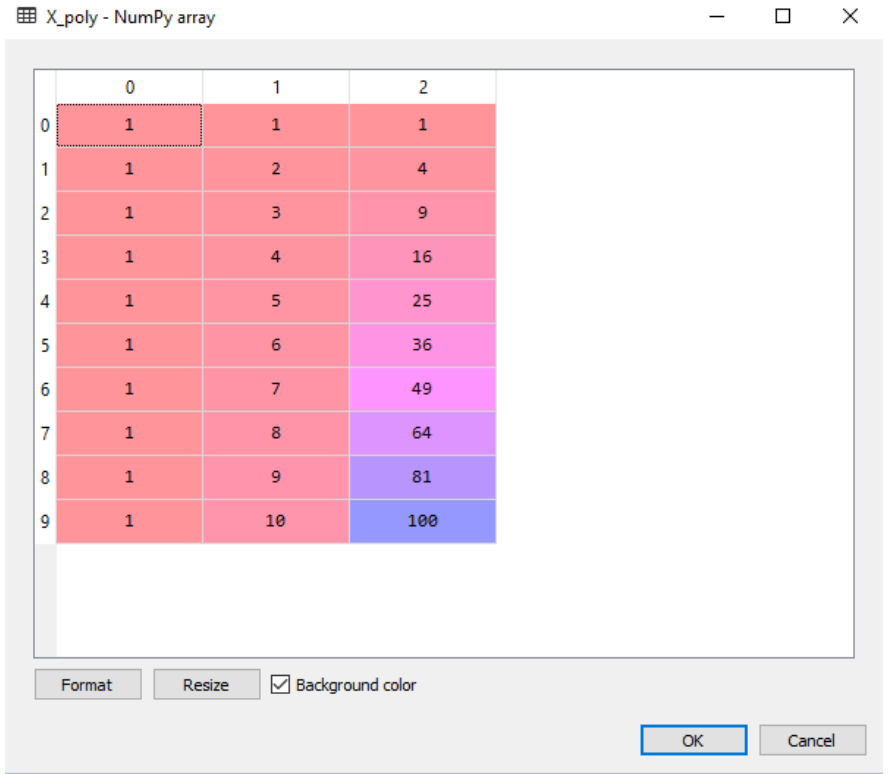
FITTING POLYNOMIAL REGRESSION

#1. We need to change X to have more independent variables:

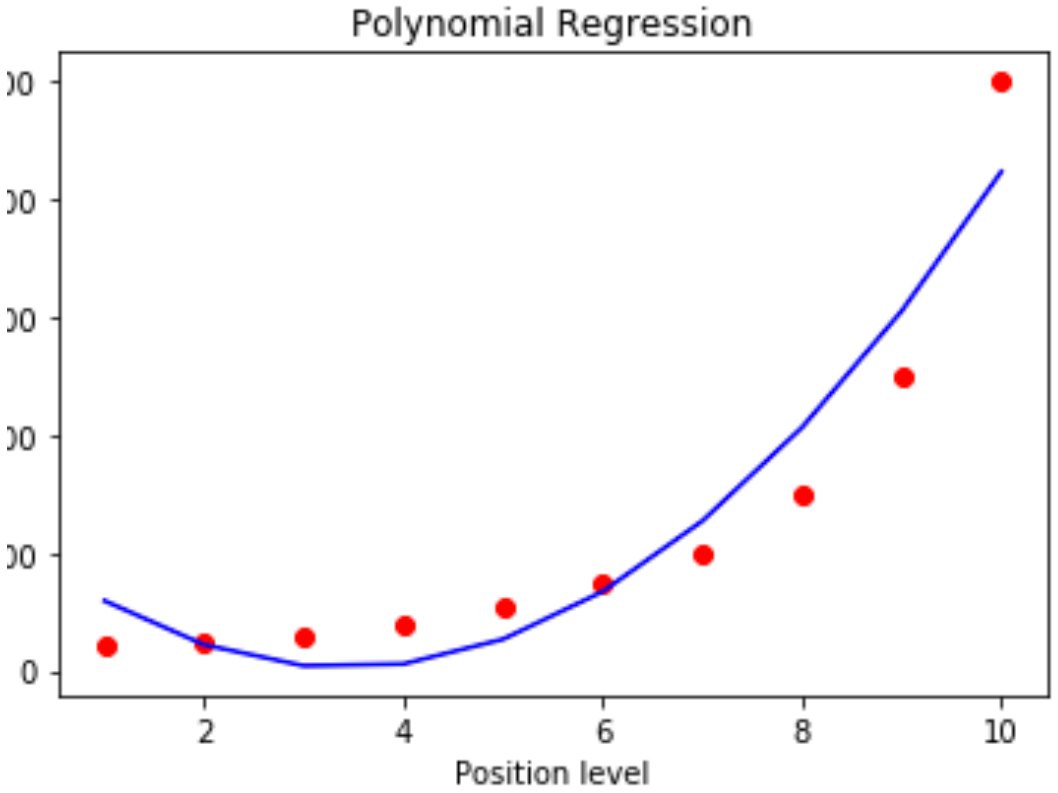
```
from sklearn.preprocessing import PolynomialFeatures

poly_reg = PolynomialFeatures(degree = 2)
X_poly = poly_reg.fit_transform(X)
```

#2. Now we can use linear regression model:

```
lin_reg_2 = LinearRegression()

lin_reg_2.fit(X_poly, y)
```



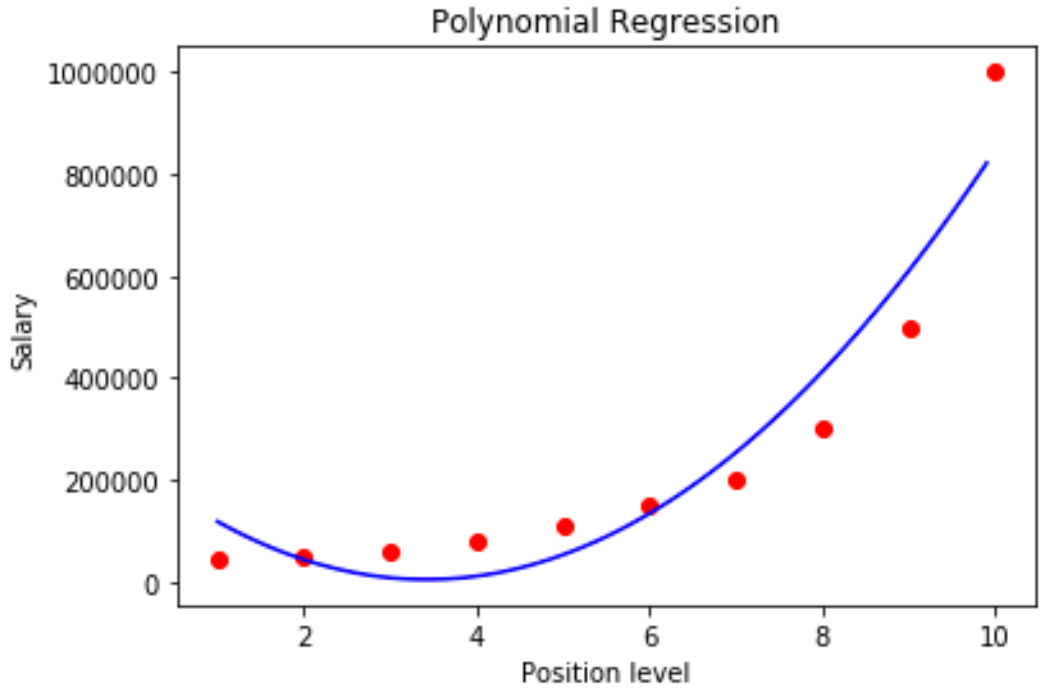
VISUALIZING THE POLYNOMIAL REGRESSION RESULTS




```
plt.scatter(X, y, color = 'red')

plt.plot(X,
lin_reg_2.predict(poly_reg.fit_transform(X))
, color = 'blue')

plt.title('Polynomial Regression')

plt.xlabel('Position level')

plt.ylabel('Salary')

plt.show()


#Nice, but can be improved.
```

SMOOTHER CURVE

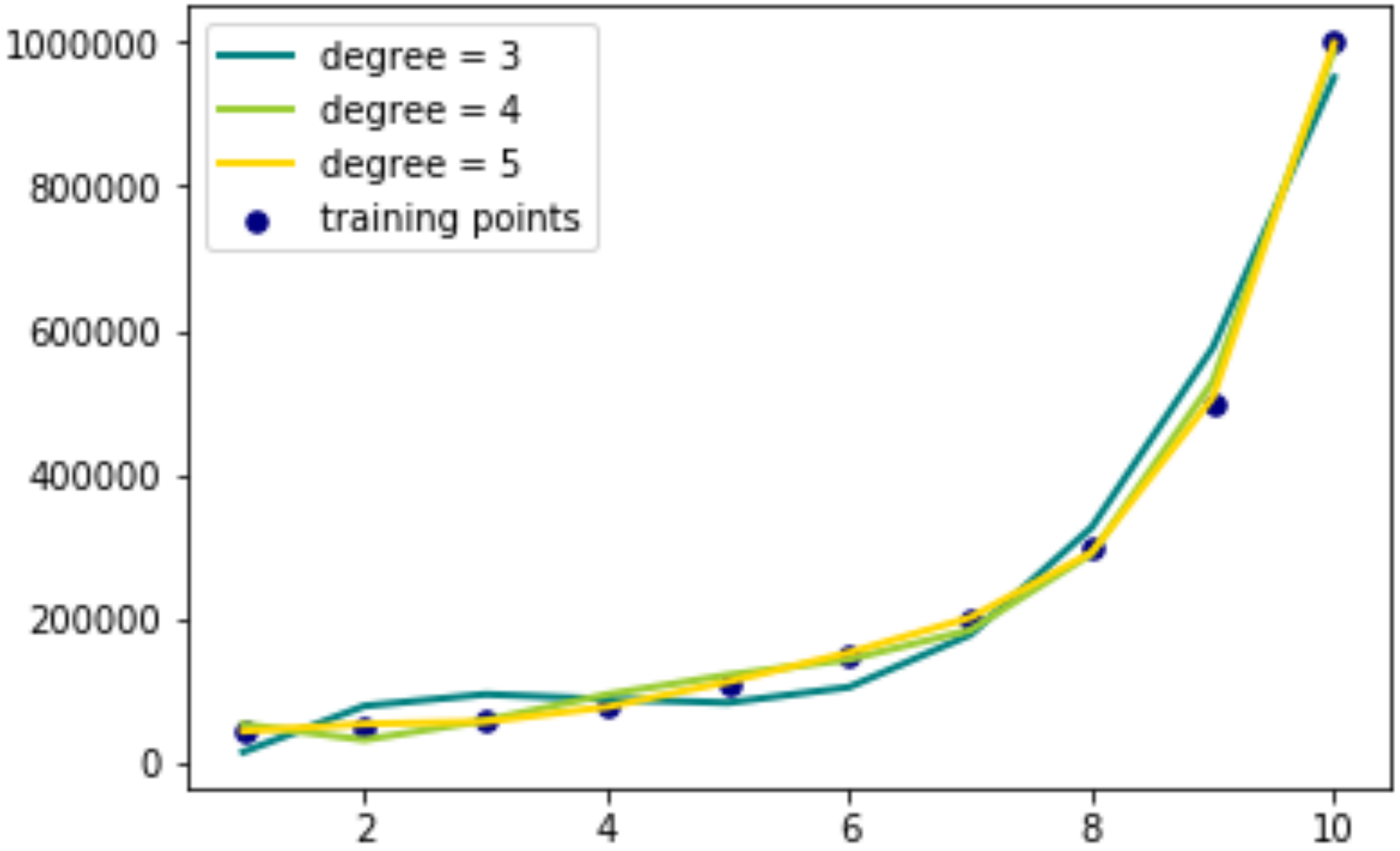
```
X_grid = np.arange(min(X), max(X), 0.1)          #we get a vector
X_grid = X_grid.reshape((len(X_grid), -1))       #we need a matrix, have single sample
plt.scatter(X, y, color = 'red')
plt.plot(X_grid, lin_reg_2.predict(poly_reg.fit_transform(X_grid)), color = 'blue')
plt.title('Polynomial Regression')
plt.xlabel('Position level')
plt.ylabel('Salary')
plt.show()
```



CROSS VALIDATION

```
from sklearn.model_selection import cross_val_score

accuracies=cross_val_score(estimator=lin_reg2,X=X_poly,y=y,cv=10,scoring='neg_
mean_squared_error')  #you can also try scoring='r2'

print(accuracies.mean())

print(accuracies.std())
```

EVALUATING MULTIPLE DEGREES...



EVALUATION

```
from sklearn.metrics import mean_squared_error, r2_score
rmse = np.sqrt(mean_squared_error(y,y_poly_pred))
r2 = r2_score(y,y_poly_pred)
```

Polynomial Regression Results:

Degree	RMSE	R^2
1	163388.74	0.67
2	82212.12	0.92
3	38931.50	0.98
4	14503.23	1.00
5	4047.50	1.00